

SQL & PL/SQL for Enterprise Development

From SQL Basics to Advanced PL/SQL with Indexing, Partitioning, PRAGMA & Oracle APEX

Looking to master Oracle SQL and PL/SQL from the ground up? This course takes you on a structured journey from fundamental query writing to advanced PL/SQL programming, culminating in a practical introduction to Oracle APEX for building data-driven applications. By the end of this course, you will be able to design efficient SQL queries, write robust PL/SQL programs, and deploy a basic Oracle APEX application.

This course is focused on:

- SQL Query Writing (Basic to Advanced)
- PL/SQL Programming (Procedures, Functions, Packages, Triggers, Collections)
- Oracle Database Object Management (DDL/DML, Indexing, Partitioning)
- PRAGMA Directives including AUTONOMOUS_TRANSACTION
- Performance Tuning and Query Optimization
- Introduction to Oracle APEX Application Development

Course Core Objectives

- SQL Fundamentals: SELECT, Filtering, Joins, Subqueries
- Aggregate & Analytical (Window) Functions
- DDL, DML & Transaction Control
- PL/SQL Block Structure, Variables, Control Flow & Loops
- Cursors, Exception Handling, Stored Procedures & Functions
- Packages, Triggers (Deep Dive), Collections (Deep Dive)
- PRAGMA AUTONOMOUS_TRANSACTION & Advanced PL/SQL Directives
- Indexing (B-tree, Bitmap, Function-based, Composite)
- Table Partitioning (Range, List, Hash, Composite, Interval)
- Bulk Processing: BULK COLLECT & FORALL
- Introduction to Oracle APEX
- Capstone Project: End-to-End Application

Target Audience

- Aspiring Oracle Database Developers
- SQL users wanting to learn PL/SQL programming
- Backend and Application Developers working with Oracle
- Data Analysts looking to automate reporting with PL/SQL

Course Pre-Requisites

- Basic computer literacy and familiarity with any operating system
- No prior SQL or programming experience required

Course Duration

- 44 Hours Total
- 22 Classes
- 2 Hours per Class

Details Course Outlines

Lesson-01

Introduction to Databases & SQL Environment

- What is a Database? (RDBMS concepts)
 - Introduction to Oracle Database
 - Overview of SQL and its role in Oracle
 - Install and configure Oracle Database (XE/Developer)
 - Introduction to SQL*Plus and SQL Developer
 - Connecting to Oracle Database
 - Understanding the HR/Scott sample schema
 - Writing and executing your first SQL query
 - SQL statement categories: DDL, DML, DCL, TCL
 - Best practices and SQL coding conventions
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Lesson-02

SQL Fundamentals – SELECT & Filtering

- The SELECT statement structure
 - Selecting all columns vs specific columns
 - Using column aliases (AS keyword)
 - Arithmetic expressions in SELECT
 - The WHERE clause – filtering rows
 - Comparison operators (=, !=, <, >, <=, >=)
 - Logical operators: AND, OR, NOT
 - BETWEEN, IN, LIKE, IS NULL operators
 - Eliminating duplicates with DISTINCT
 - Ordering results with ORDER BY (ASC/DESC)
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Lesson-03

SQL Single-Row Functions

- Introduction to Oracle built-in functions
 - Character functions: UPPER, LOWER, INITCAP, LENGTH
 - String manipulation: SUBSTR, INSTR, CONCAT, REPLACE
 - Number functions: ROUND, TRUNC, MOD, ABS
 - Date functions: SYSDATE, ADD_MONTHS, MONTHS_BETWEEN
 - Date formatting with TO_CHAR and TO_DATE
 - Conversion functions: TO_NUMBER, CAST
 - NVL, NVL2, NULLIF, COALESCE for null handling
 - DECODE and CASE expressions
 - Nesting functions (function inside function)
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Lesson-04

Aggregation & Grouping (Group Functions)

- Introduction to aggregate/group functions
 - COUNT, SUM, AVG, MIN, MAX
 - Counting rows vs non-null values
 - GROUP BY clause – grouping data
 - Filtering groups with HAVING
 - Difference between WHERE and HAVING
 - Grouping on multiple columns
 - GROUPING SETS for flexible aggregation
 - Combining group functions with single-row functions
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Lesson-05

Joining Tables

- Why joins are needed (relational model recap)
 - INNER JOIN – matching rows from both tables
 - LEFT OUTER JOIN and RIGHT OUTER JOIN
 - FULL OUTER JOIN
 - CROSS JOIN and Cartesian products
 - SELF JOIN – joining a table to itself
 - Joining more than two tables
 - Using table aliases for readability
 - Troubleshooting common join issues
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Lesson-06

Subqueries

- Introduction to subqueries (nested queries)
 - Single-row subqueries with =, !=, <, >
 - Multiple-row subqueries with IN, ANY, ALL
 - Correlated subqueries – reference to outer query
 - EXISTS and NOT EXISTS operators
 - Subqueries in the FROM clause (inline views)
 - Subqueries in the SELECT clause (scalar subqueries)
 - Nested subqueries (multiple levels deep)
 - Using subqueries for data validation
 - Performance considerations for subqueries
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Lesson-07

DDL – Creating and Managing Database Objects

- Introduction to Data Definition Language (DDL)
 - CREATE TABLE – column definitions and data types
 - Oracle data types: VARCHAR2, NUMBER, DATE, CLOB, BLOB
 - Constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK
 - ALTER TABLE – add, modify, drop columns and constraints
 - Renaming and dropping tables (RENAME, DROP)
 - TRUNCATE vs DELETE – differences and use cases
 - Temporary tables in Oracle
 - Creating and managing SEQUENCES
 - Creating and using SYNONYMS and VIEWS
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Lesson-08

DML, TCL & Data Integrity

- INSERT statement – single and multiple row inserts
- INSERT INTO ... SELECT for bulk inserts
- UPDATE statement – modifying existing data
- DELETE statement – removing rows
- MERGE statement (UPSERT pattern)
- Transaction Control Language: COMMIT and ROLLBACK
- SAVEPOINT and partial rollbacks
- Implicit vs explicit transactions in Oracle
- Understanding data integrity with constraints
- Handling constraint violations gracefully

Lesson-09

Advanced SQL – Analytical & Window Functions

- Introduction to analytic (window) functions
- ROW_NUMBER(), RANK(), DENSE_RANK()
- NTILE() for data distribution buckets
- LAG() and LEAD() for accessing prior/next row values
- FIRST_VALUE() and LAST_VALUE()
- SUM/AVG/COUNT with OVER() and PARTITION BY
- Framing with ROWS BETWEEN and RANGE BETWEEN
- Common Table Expressions (CTEs) with WITH clause
- Recursive CTEs for hierarchical data
- PIVOT and UNPIVOT for data transformation

Lesson-10

Introduction to PL/SQL – Basics & Block Structure

- What is PL/SQL and why use it?
- PL/SQL block structure: DECLARE, BEGIN, EXCEPTION, END
- Anonymous blocks vs named blocks
- Variables: declaration, data types, and initialization
- %TYPE and %ROWTYPE attribute declarations
- Constants and default values
- Scalar data types: VARCHAR2, NUMBER, BOOLEAN, DATE
- Composite types overview
- Outputting data with DBMS_OUTPUT.PUT_LINE
- Writing, running and debugging anonymous blocks

Lesson-11

PL/SQL Control Structures

- Conditional logic: IF-THEN-ELSIF-ELSE
- CASE statements and CASE expressions in PL/SQL
- Basic LOOP with EXIT and EXIT WHEN
- WHILE LOOP structure and use cases
- FOR LOOP (numeric) and its implicit counter
- Nested loops and loop labels

- CONTINUE statement for skipping iterations
 - GOTO statement (and when NOT to use it)
 - NULL statement as a placeholder
 - Practical exercises combining control structures
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Lesson-12

PL/SQL Cursors

- What is a cursor? Implicit vs explicit cursors
 - Implicit cursor attributes: %FOUND, %NOTFOUND, %ROWCOUNT
 - Declaring and opening explicit cursors
 - FETCH INTO and closing cursors
 - Cursor FOR LOOP (simplified cursor handling)
 - Parameterized cursors
 - Cursor variables (REF CURSOR)
 - SYS_REFCURSOR – weakly typed REF CURSOR
 - Using cursors to process bulk data sets
 - Cursor best practices and memory management
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Lesson-13

PL/SQL Exception Handling

- What is an exception? Predefined vs user-defined
 - Common predefined exceptions: NO_DATA_FOUND, TOO_MANY_ROWS
 - ZERO_DIVIDE, VALUE_ERROR, DUP_VAL_ON_INDEX
 - WHEN OTHERS – catch-all handler
 - SQLCODE and SQLERRM for error details
 - Declaring and raising user-defined exceptions (RAISE)
 - RAISE_APPLICATION_ERROR for custom error messages
 - Exception propagation across nested blocks
 - Re-raising exceptions
 - Best practices for robust exception handling
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Lesson-14

Stored Procedures & Functions

- Difference between procedures and functions
- CREATE OR REPLACE PROCEDURE syntax

- IN, OUT, and IN OUT parameter modes
 - Calling procedures from SQL*Plus, SQL Developer, and PL/SQL
 - CREATE OR REPLACE FUNCTION syntax
 - Functions that return scalar values
 - Functions returning REF CURSOR
 - Using functions in SQL SELECT statements
 - Dropping and recompiling subprograms
 - Viewing errors with SHOW ERRORS / USER_ERRORS
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Lesson-15

PL/SQL Packages

- What is a package? Package spec vs package body
 - Benefits of packages: encapsulation and performance
 - Creating a package specification
 - Creating a package body
 - Public vs private procedures and variables
 - Package-level (global) variables and constants
 - Overloading subprograms within a package
 - Package initialization section
 - Oracle built-in packages overview: DBMS_OUTPUT, DBMS_SCHEDULER, UTL_FILE
 - Designing reusable package libraries
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Lesson-16

Database Triggers (Deep Dive)

- What is a trigger and when should you use one?
 - DML triggers: BEFORE INSERT, AFTER INSERT, BEFORE UPDATE, AFTER DELETE
 - Row-level triggers (:NEW and :OLD pseudo-records)
 - Statement-level triggers – when row context is not needed
 - WHEN clause for conditional trigger firing
 - Compound triggers – combining timing points in one object (Oracle 11g+)
 - INSTEAD OF triggers on views for non-updatable view DML
 - DDL triggers – auditing schema changes (CREATE, ALTER, DROP events)
 - System/database event triggers (LOGON, LOGOFF, SERVERERROR)
 - Trigger best practices: avoiding mutating table errors, keeping logic minimal
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Lesson-17

PL/SQL Collections (Deep Dive)

- ▶ Three collection types: Associative Arrays, Nested Tables, VARRAYs – comparison
- ▶ Associative Arrays (INDEX BY): VARCHAR2 and PLS_INTEGER keys
- ▶ Nested Tables: declaration, initialization with constructor, storage in DB
- ▶ VARRAYs: fixed maximum size, dense storage, use cases
- ▶ Collection methods: COUNT, FIRST, LAST, NEXT, PRIOR, EXISTS, EXTEND, TRIM, DELETE
- ▶ Iterating collections: numeric FOR loop vs WHILE with FIRST/NEXT
- ▶ Bulk loading into collections using BULK COLLECT INTO
- ▶ FORALL statement – bulk DML using collection data
- ▶ Passing collections as IN/OUT parameters to procedures
- ▶ Nested collections (collection of collections) and practical patterns

Lesson-18

Indexing & Performance Tuning

- ▶ What is an index and how does Oracle use it? (B-tree internals)
- ▶ B-tree indexes: CREATE INDEX, UNIQUE INDEX, COMPOSITE INDEX
- ▶ Bitmap indexes: when to use them (low-cardinality columns, DW workloads)
- ▶ Function-based indexes: indexing on expressions and UPPER(column)
- ▶ Invisible indexes – testing impact without affecting the optimizer
- ▶ Index selectivity: high vs low cardinality and optimizer decisions
- ▶ Reading EXPLAIN PLAN: Full Table Scan vs Index Range Scan vs Index Unique Scan
- ▶ AUTOTRACE and DBMS_XPLAN.DISPLAY for execution plan analysis
- ▶ Optimizer hints: /*+ INDEX */ /*+ FULL */ /*+ USE_NL */ /*+ PARALLEL */
- ▶ Index maintenance: REBUILD, COALESCE, monitoring usage with V\$OBJECT_USAGE

Lesson-19

Table Partitioning

- ▶ Why partition? Benefits for large tables (performance, manageability, archiving)
- ▶ Range Partitioning: partitioning by date ranges (most common pattern)
- ▶ List Partitioning: partitioning by discrete values (region, status, category)
- ▶ Hash Partitioning: even data distribution across N partitions
- ▶ Composite Partitioning: Range-List, Range-Hash combinations
- ▶ Interval Partitioning: automatic partition creation for date/number ranges
- ▶ Partition Pruning: how Oracle eliminates irrelevant partitions at query time
- ▶ Local and Global indexes on partitioned tables
- ▶ Partition maintenance: ADD, DROP, MERGE, SPLIT, EXCHANGE PARTITION
- ▶ Querying partition metadata: USER_TAB_PARTITIONS, USER_IND_PARTITIONS

Lesson-20

PRAGMA AUTONOMOUS_TRANSACTION & Advanced PL/SQL

- What is a PRAGMA in PL/SQL? Compiler directives overview
- PRAGMA AUTONOMOUS_TRANSACTION – definition and real-world use case
- How autonomous transactions run in their own independent transaction scope
- Classic use case: writing audit/error logs that survive a ROLLBACK
- Creating an autonomous logging procedure step-by-step
- Calling autonomous procedures from triggers and main transactions
- PRAGMA EXCEPTION_INIT – binding Oracle error codes to named exceptions
- PRAGMA SERIALLY_REUSABLE – controlling package-level state in stateless sessions
- PRAGMA RESTRICT_REFERENCES – purity constraints on functions called from SQL
- Common pitfalls with PRAGMA AUTONOMOUS_TRANSACTION (commit discipline, deadlocks)

Lesson-21

Introduction to Oracle APEX

- What is Oracle Application Express (APEX)?
- APEX architecture and how it runs on Oracle DB
- Accessing APEX: workspace setup and login
- Overview of the APEX development environment
- Creating your first APEX application
- Building a simple Interactive Report from a SQL query
- Creating a basic Form page for data entry
- Using the APEX SQL Workshop for ad-hoc queries
- Introduction to APEX page processes and validations
- Next steps: resources for continued APEX learning

Lesson-22

Capstone Project & Course Review

- Capstone project overview: design a mini HR/Payroll schema
- Create normalized tables with PRIMARY KEY, FOREIGN KEY, CHECK constraints
- Write complex multi-table queries with JOINS and analytical functions
- Build a PL/SQL package: procedures, functions, cursors, and exception handling
- Implement a PRAGMA AUTONOMOUS_TRANSACTION audit log triggered on DML
- Apply partitioning to the largest table and verify partition pruning

- Create indexes and compare EXPLAIN PLAN before and after
 - Bulk data load simulation using BULK COLLECT / FORALL
 - Deploy a simple APEX Interactive Report and Form on the project schema
 - Course summary, Q&A, certification roadmap (OCA/OCP 1Z0-071 / 1Z0-149)
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